3(2) $\begin{pmatrix}-1&0&4\\3&0&-1\\0&1&-1\end{pmatrix} \Rightarrow \begin{pmatrix}-1&0&0\\3&0&11\\0&1&0\end{pmatrix} \Rightarrow \begin{pmatrix}-1&0&0\\0&0&11\\0&1&0\end{pmatrix} \Rightarrow \begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix}$

$\begin{pmatrix}1&-1&5\\-1&4&-2\\0&3&3\end{pmatrix} \Rightarrow \begin{pmatrix}0&0&0\\-1&4&-2\\0&3&3\end{pmatrix} \Rightarrow \begin{pmatrix}0&0&0\\-1&40&-20\\0&3&0\end{pmatrix} \Rightarrow \begin{pmatrix}1&0&0\\0&1&0\\0&0&0\end{pmatrix}$

故二者不没有相同的相抵标准形

4. 充分性：设 $P_i Q_i$ 为初等矩阵，设 A 的基本相抵形为 A_r 存在 P_i, Q_i 使

∵ 若 $|A|$ 非异 ∴ $|A| \neq 0$ ∴ $A = P_1 P_2 \cdots P_k A_r Q_1 Q_2 \cdots Q_j$

若 $A_r \neq I_n$ 则 $A_r = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix} (r<n)$ ∴ $|A_r| = 0$

∴ $|A| = 0$，与条件矛盾 ∴ $A_r = I_n$ 即 A 与 I_n 相抵

必要性：∵ I_n 与 A 相抵 存在 $P_1 \cdots P_i Q_1 \cdots Q_j$ 使 ∴ $A = P_1 P_2 \cdots P_i Q_1 Q_2 \cdots Q_j$

∴ $|A| = |P_1||P_2| \cdots |P_i||Q_1| \cdots |Q_j| \neq 0$

∴ A 非异

5. ~~取 m 阶可逆矩阵 P_1 与 n 阶可逆阵 Q_1~~

~~则~~

5. 由定理 2.4.1 知 $A_{m\times n}$ 必相抵于一标准相抵形，设其为 B，则存在 $P_1 \cdots P_i Q_1 \cdots Q_j$ 使得

则 $B = P_1 P_2 \cdots P_i A Q_1 Q_2 \cdots Q_j$ ($P_i Q_i$ 为初等矩阵)

设 $P = P_1 P_2 \cdots P_i$ $Q = Q_1 Q_2 \cdots Q_j$ ∴ $|P| = |P_1| \cdots |P_i| \neq 0$ $|Q| = |Q_1| \cdots |Q_j| \neq 0$

∴ P, Q 为可逆阵 ∴ PAQ 存在 P, Q 使 PAQ 是其标准相抵型

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2.5.1(3) $\begin{pmatrix} 1 & 2 & -1 & 1 & & \\ 3 & 1 & 0 & & 1 & \\ 7 & 6 & -2 & & & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & 2 & -1 & 1 & & \\ 3 & 1 & 0 & & 1 & \\ 4 & 0 & 1 & -3 & & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} -5 & 0 & 0 & -2 & -2 & 1 \\ 3 & 1 & 0 & -0 & 1 & 0 \\ 4 & 0 & 1 & -3 & 0 & 1 \end{pmatrix}^{-1}$

$\Rightarrow \begin{pmatrix} 1 & 0 & 0 & 2 & 2 & -1 \\ 0 & 1 & 0 & -6 & -5 & 3 \\ 0 & 0 & 1 & -11 & -8 & 5 \end{pmatrix}$

∴逆阵为 $\begin{pmatrix} 2 & 2 & -1 \\ -6 & -5 & 3 \\ -11 & -8 & 5 \end{pmatrix}$

2. $(A\ I_n) = \begin{pmatrix} a_1 & & & & & 1 & & \\ & a_2 & & & & & \ddots & \\ & & \ddots & & & & & 1 \\ & & & a_{n-1} & & & & \\ a_n & & & & & & & \end{pmatrix} = \begin{pmatrix} 1 & & & & \frac{1}{a_n} \\ & \ddots & & & \\ & & \frac{1}{a_1} & & \\ & & & \ddots & \\ & & & & \frac{1}{a_{n-1}} \end{pmatrix}$

∴A的逆阵为 $\begin{pmatrix} & & & \frac{1}{a_n} \\ \frac{1}{a_1} & & & \\ & \frac{1}{a_2} & & \\ & & \ddots & \\ & & \frac{1}{a_{n-1}} & \end{pmatrix}$

3. $X = \begin{pmatrix} 2 & 2 & 3 \\ 1 & -1 & 0 \\ -1 & 2 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 4 & -1 \\ 2 & 1 \\ 1 & 0 \end{pmatrix}$

$\begin{pmatrix} 2 & 2 & 3 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 & 1 & 0 \\ -1 & 2 & 1 & 0 & 0 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & 0 & 0 & 1 & -4 & -3 \\ 1 & -1 & 0 & & 1 & \\ -1 & 2 & 1 & & & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & 0 & 0 & 1 & -4 & -3 \\ 1 & -1 & 0 & & 1 & \\ -1 & 0 & 1 & & 2 & 1 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} 1 & & & 1 & -4 & -3 \\ & 1 & & 1 & -5 & -3 \\ & & 1 & -1 & 6 & 4 \end{pmatrix}$

$\therefore \begin{pmatrix} 2 & 2 & 3 \\ 1 & -1 & 0 \\ -1 & 2 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & -4 & -3 \\ 1 & -5 & -3 \\ -1 & 6 & 4 \end{pmatrix}$

$\therefore X = \begin{pmatrix} 1 & -4 & -3 \\ 1 & -5 & -3 \\ -1 & 6 & 4 \end{pmatrix} \begin{pmatrix} 4 & -1 \\ 2 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -7 & -5 \\ -9 & -6 \\ 12 & 7 \end{pmatrix}$

5. $S = \begin{pmatrix} 1 & 1 & & 1 \\ x_1 & x_2 & \cdots & x_n \\ & & & \\ x_1^{n-1} & x_2^{n-1} & & x_n^{n-1} \end{pmatrix} \times \begin{pmatrix} 1 & x_1 & & x_1^{n-1} \\ 1 & x_2 & & x_2^{n-1} \\ & & & \\ 1 & x_n & & x_n^{n-1} \end{pmatrix}$ 转

对应两者行列式值为 $\prod_{1 \le i < j \le n} (x_j - x_i)$

$\therefore |S| = \left[\prod_{1 \le i < j \le n} (x_j - x_i)\right]^2 \ge 0$

6. 取矩阵 $F = \begin{pmatrix} 1 & 1 & 1 & & 1 \\ 1 & \varepsilon & \varepsilon^2 & \cdots & \varepsilon^{n-1} \\ 1 & \varepsilon^2 & \varepsilon^4 & & \varepsilon^{2(n-1)} \\ & & & & \\ 1 & \varepsilon^{n-1} & \varepsilon^{2(n-1)} & & \varepsilon^{(n-1)^2} \end{pmatrix}$ $F = \begin{pmatrix} 1 & 1 & 1 & & 1 \\ 1 & w_n^1 & w_n^2 & \cdots & w_n^{n-1} \\ 1 & w_n^2 & w_n^4 & & w_n^{2(n-1)} \\ & & & & \\ 1 & w_n^{n-1} & w_n^{2(n-1)} & & w_n^{(n-1)^2} \end{pmatrix}$ 为 $w_n = \cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n}$

$\therefore |F| = n \quad F^{-1} = \frac{F}{n}$

$\therefore F^{-1}AF = \begin{pmatrix} f(\varepsilon_1) & & & \\ & f(\varepsilon_2) & & \\ & & \ddots & \\ & & & f(\varepsilon_n) \end{pmatrix}$ $\therefore |F^{-1}AF| = |A| = f(\varepsilon_1) f(\varepsilon_2) \cdots f(\varepsilon_n)$

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7. 由6. $|A_n| = f(\varepsilon_1) f(\varepsilon_2) \cdots f(\varepsilon_n)$

设 $w_n = \cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n}$

当 $i = n$ 时 $f(\varepsilon_i) = n$

$i \neq n$ 时 $f(\varepsilon_i) = 1 + 2w_i + 3w_i^2 + \cdots + nw_i^{n-1}$

$= \frac{(nw_i^{n-1})(1-w_i) - (1-w_i^n)(-1)}{(1-w_i)^2}$

$\therefore |A_n| = n \prod_{i=1}^{n-1} \frac{(nw_i^{n-1})(1-w_i) + (1-w_i^n)}{(1-w_i)^2}$

2.b 1(2) 设左为 $\begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$ 右为 $\begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix}$ 积为 $C = \begin{pmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{pmatrix}$

$C_{11} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ $C_{12} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ -1 & 1 \end{pmatrix}$

$C_{21} = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & -1 \end{pmatrix}$ $C_{22} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$

$\therefore C = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & -1 & 0 & 2 \end{pmatrix}$

2(2) $\begin{pmatrix} A_{11}B_{11} & A_{11}B_{12}+A_{12}B_{22} & A_{11}B_{13}+A_{12}B_{23}+A_{13}B_{33} \\ 0 & A_{22}B_{22} & A_{22}B_{23}+A_{23}B_{33} \\ 0 & 0 & A_{33}B_{33} \end{pmatrix}$

3. 对A $A' = \begin{pmatrix} & I_{n-1} \\ I_1 & \end{pmatrix}$

设对 $k=i$ $(1<i\leq n-1)$ 时 $A^k = \begin{pmatrix} 0 & I_{n-i} \\ I_i & 0 \end{pmatrix}$

$\therefore k=i+1$ 时 $A^k = A^i A = \begin{pmatrix} 0 & I_{n-i} \\ I_i & 0 \end{pmatrix}\begin{pmatrix} 0 & I_{n-1} \\ I_1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & I_1 & 0 \\ 0 & 0 & I_{n-i-1} \\ I_i & 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & I_{n-1} \\ I_1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & I_{n-i-1} \\ I_{i+1} & 0 \end{pmatrix}$

由数学归纳法 $A^k = \begin{pmatrix} 0 & I_{n-k} \\ I_k & 0 \end{pmatrix}$

4 对A $A' = \begin{pmatrix} 0 & I_{n-1} \\ 0 & 0 \end{pmatrix}$

设 $k=i$ 时 $(1<i\leq n-1)$ $A^k = \begin{pmatrix} 0 & I_{n-i} \\ 0 & 0 \end{pmatrix}$

当 $k=i+1$ 时 ~~$A^k = \begin{pmatrix} 0 & I_{n-i} \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & I_{n-1} \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & I_1 & 0 \\ 0 & 0 & I_{n-i-1} \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & I_{n-1} \\ 0 & 0 \end{pmatrix}$~~

~~$A^k = \begin{pmatrix} 0 & I_{n-i} \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & I_{n-i} \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & I_i & 0 \\ 0 & 0 & I_{n-i-1} \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & I_{n-i} \\ 0 & 0 \end{pmatrix}$~~

$\therefore A^k = \begin{pmatrix} 0 & I_{n-i} \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & I_i & 0 \\ 0 & 0 & I_{n-i-1} \\ 0 & 0 & 0 \end{pmatrix} = 0 \times (0\ I_i\ 0) + \begin{pmatrix} I_{n-i} \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 & I_{n-i-1} \\ 0 & 0 \end{pmatrix}$

$= \begin{pmatrix} 0 & I_{n-i-1} \\ 0 & 0 \end{pmatrix}$

由数学归纳法 $A^k = \begin{pmatrix} 0 & I_{n-k} \\ 0 & 0 \end{pmatrix}$

7. $(C\ I) = \begin{pmatrix} 0 & A & I & 0 \\ B & 0 & 0 & I \end{pmatrix} = \begin{pmatrix} B & 0 & 0 & I \\ 0 & A & I & 0 \end{pmatrix} = \begin{pmatrix} I & 0 & 0 & B^{-1} \\ 0 & I & A^{-1} & 0 \end{pmatrix}$

$\therefore C^{-1} = \begin{pmatrix} 0 & B^{-1} \\ A^{-1} & 0 \end{pmatrix}$

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